meeting-2026-10-22

2026-10-22

globin enhancer and promoter positive controls on gasperini

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Experiment: compute p-values for the 130 globin control pairs in Gasperini for different assignments. These pairs are from 10 guide groups (enhancers HS1 to HS4 and globin promoters) tested against each of the 13 genes they were tested against in the original paper. We focus on the 18 pairs formed by the 6 guide groups targeting enhancers, and the genes HBE1, HBG1, and HBG2.

Assignment methods:

1. fishash
2. fishash+
3. thresholding with a threshold of 3, 5, and 10

This was done with and without cellwise QC (using SCEPTRE default if QC was done). No pairwise QC was done.

First we compare the fishash+ p-values with those from fishash or thresholding at 5. The promoter guide groups are removed, so 78 pairs remain.

This suggests that fishash+ adds power.

The same comparison on `z_orig`, the observed test statistic before sceptre compares it with its fitted skew-normal null.

The promoter guide groups are still removed (78 enhancer pairs: 18 highlighted, 60 other). There is clear evidence that fishash+ improves over thresholding, and a suggestion that it improves over fishash.

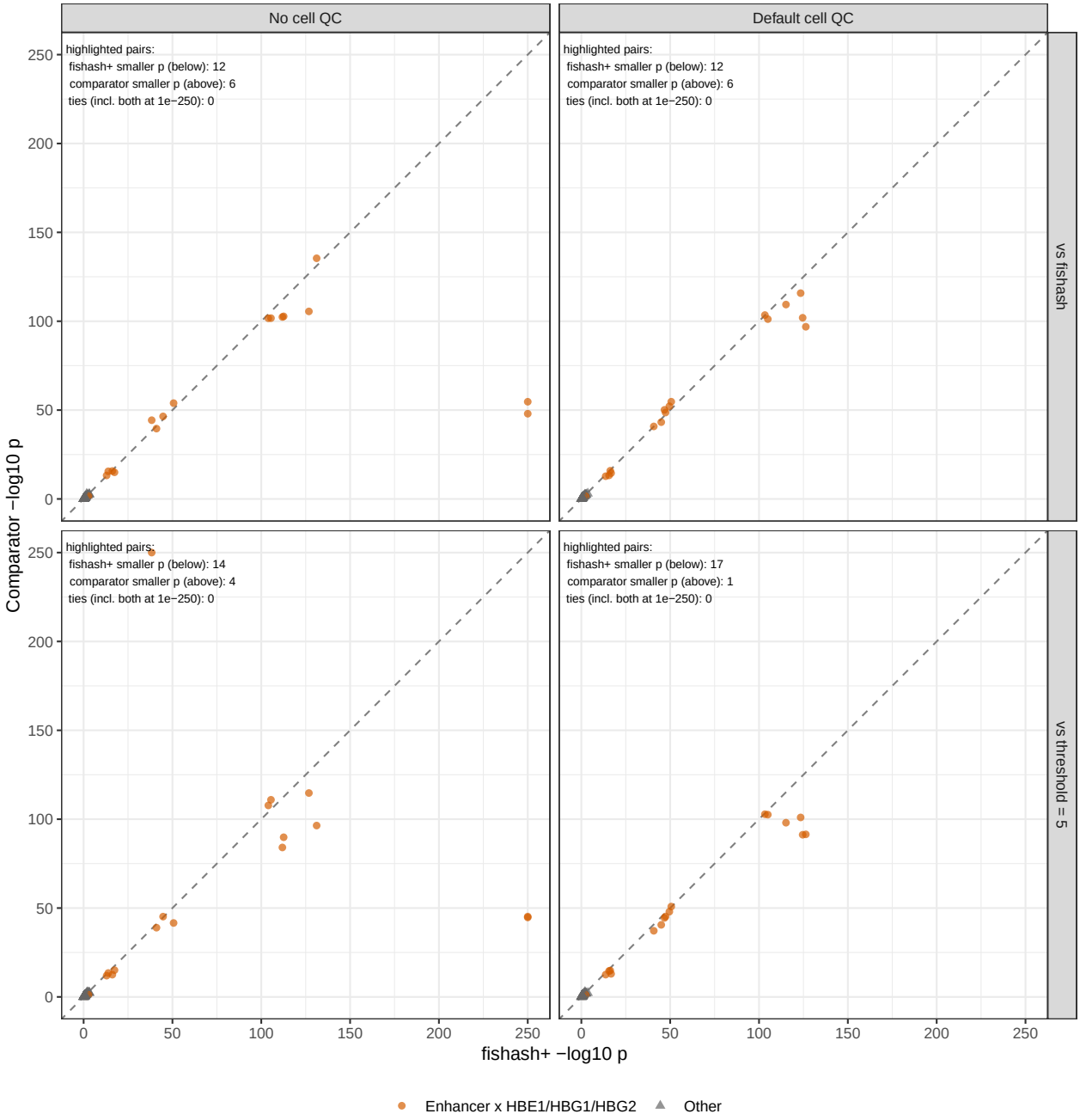
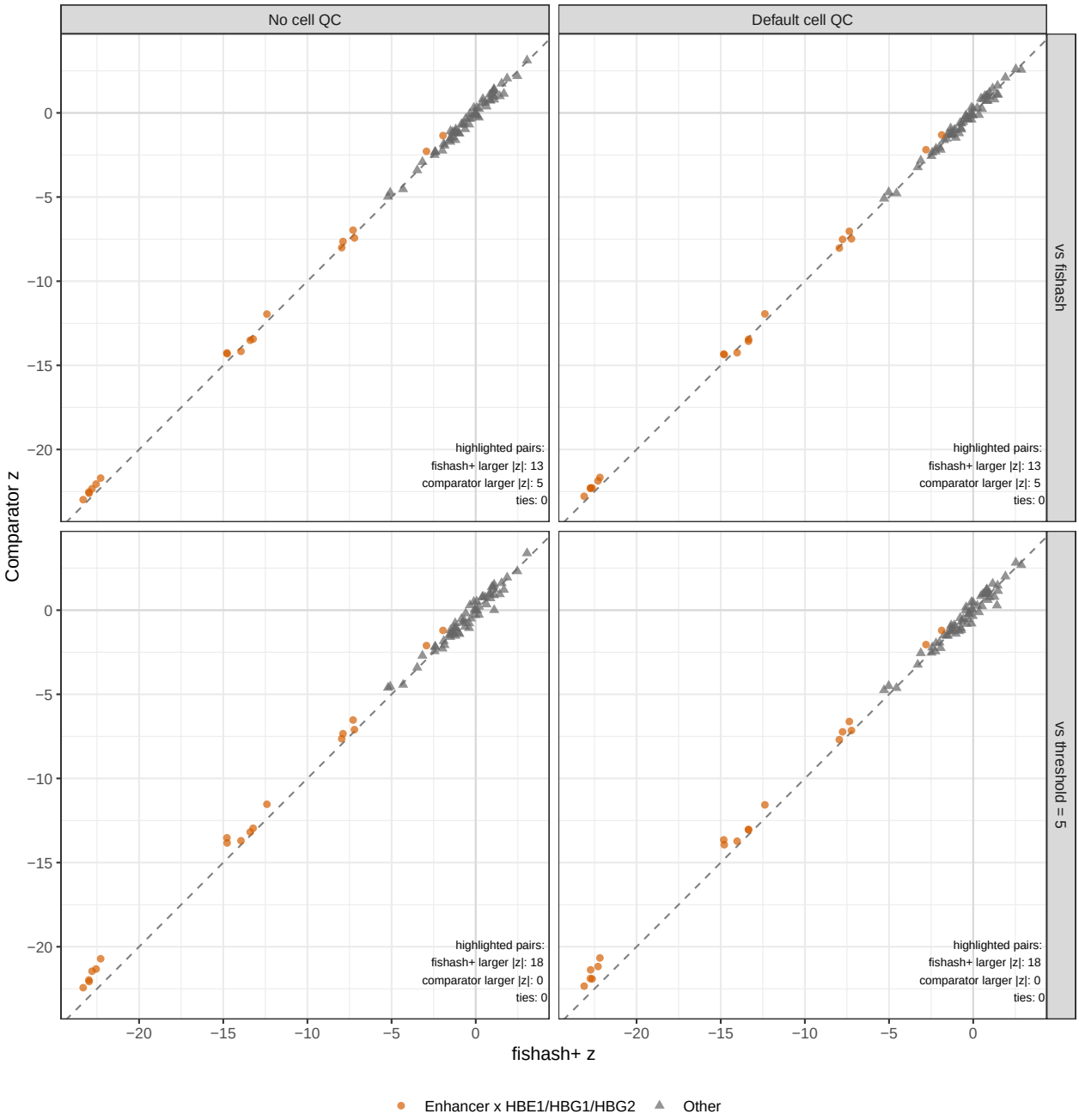


Figure 1: Only the 78 enhancer-targeting pairs. Fishash+ appears to slightly improve over fishash in this tiny sample, with stronger evidence that it improves over thresholding.



What are the extra cells fishash+ assigns? For each of the 9 enhancer-targeting guides, we show the UMI counts for (guide, cell) pairs that fishash+ called perturbed, but fishash did not. They are indeed all small, with 2 being the universal mode. Note that every guide-cell pair that fishash assigns is also assigned by fishash+, so the converse plot is empty.

So fishash+ is adding a lot of cells with tiny UMI counts. Do those help or hurt power?

```
##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

##
```

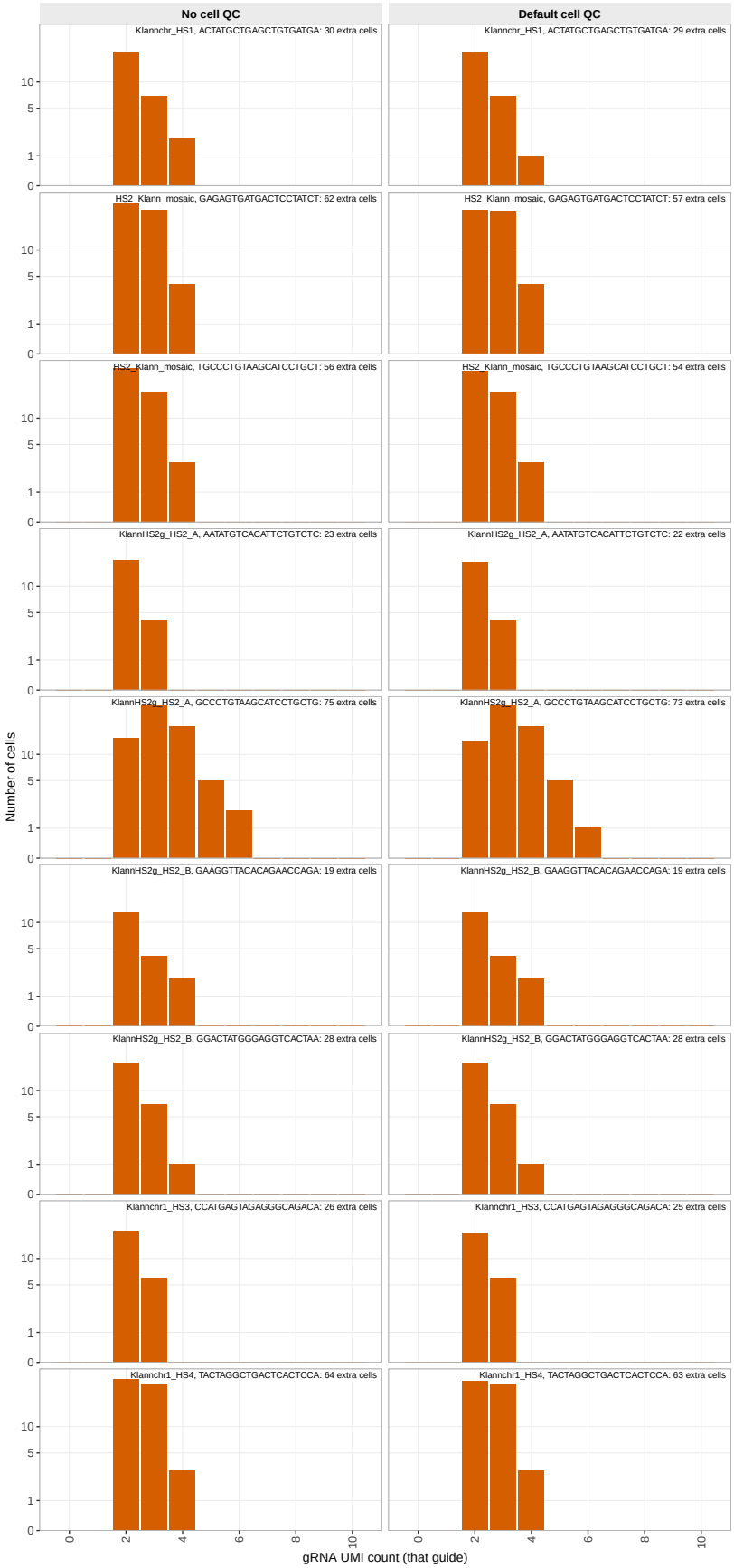
```
## Attaching package: 'data.table'

## The following objects are masked from 'package:lubridate':
##
##     hour, isoweek, isoyear, mday, minute, month, quarter, second, wday,
##     week, yday, year

## The following objects are masked from 'package:dplyr':
##
##     between, first, last

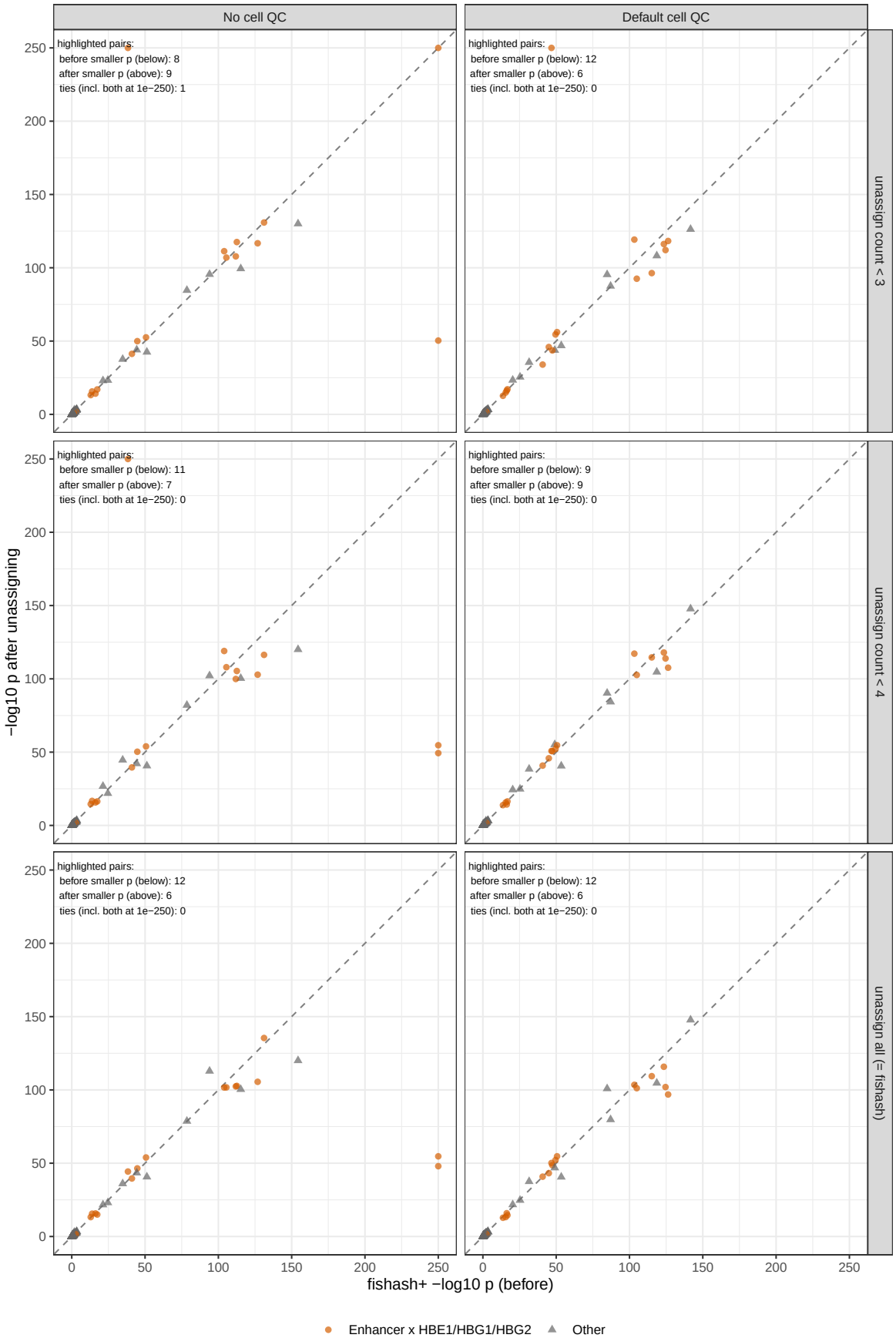
## The following object is masked from 'package:purrr':
##
##     transpose

## The following object is masked from 'package:base':
##
##     %notin%
```

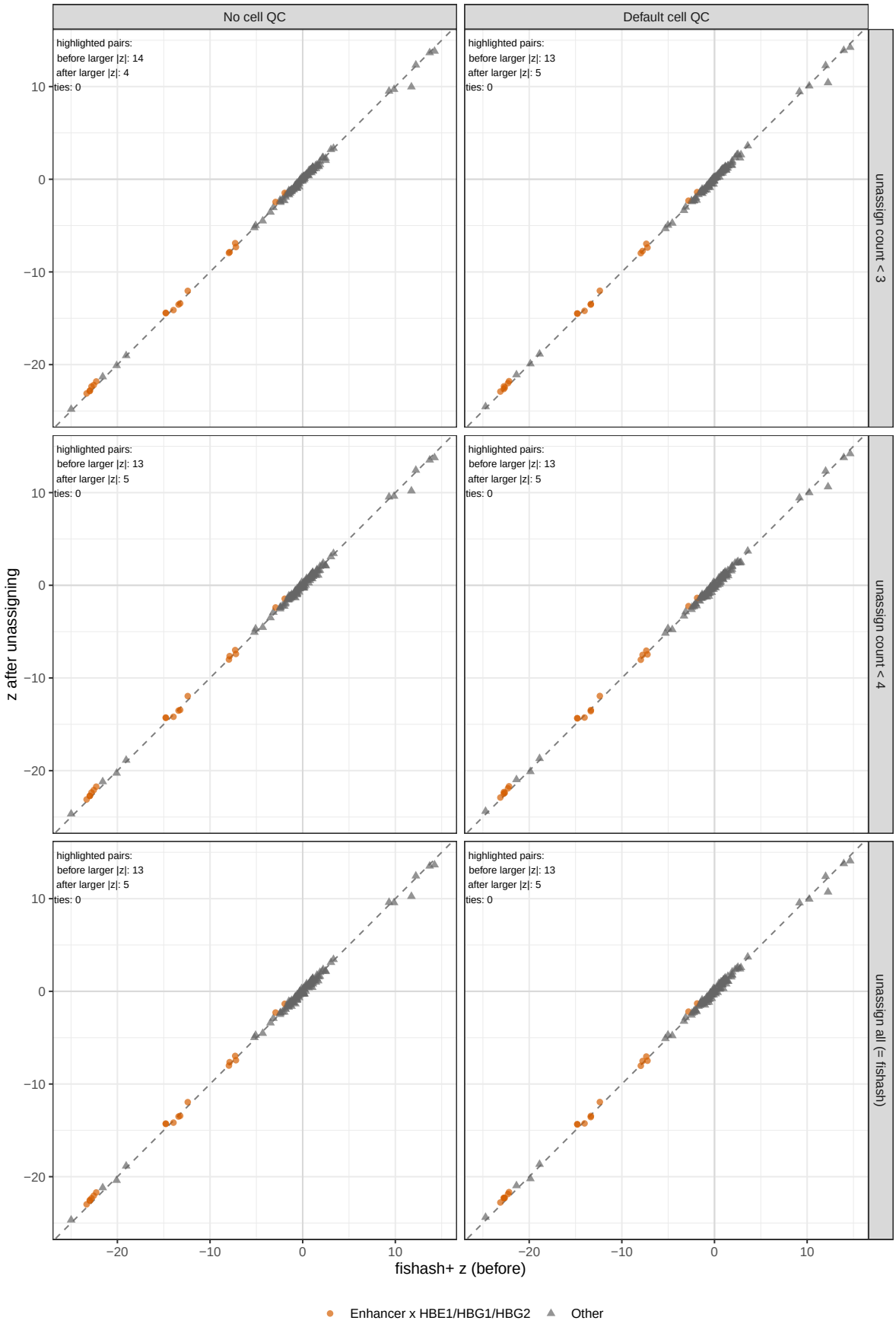

No cell QC
Default cell QC
Klannchr_HS1, ACTATGCTGAGCTGTGATGA: 30 extra cells
Klannchr_HS1, ACTATGCTGAGCTGTGATGA: 29 extra cells
HS2_Klann_mosaic, GAGAGTGATGACTCCTATCT: 62 extra cells
HS2_Klann_mosaic, GAGAGTGATGACTCCTATCT: 57 extra cells
HS2_Klann_mosaic, TGCCCTGTAAGCATCCTGCT: 56 extra cells
HS2_Klann_mosaic, TGCCCTGTAAGCATCCTGCT: 54 extra cells
KlannHS2g_HS2_A, AATATGTCACATTCTGTCTC: 23 extra cells
KlannHS2g_HS2_A, AATATGTCACATTCTGTCTC: 22 extra cells
KlannHS2g_HS2_A, GCCCTGTAAGCATCCTGCTG: 75 extra cells
KlannHS2g_HS2_A, GCCCTGTAAGCATCCTGCTG: 73 extra cells
KlannHS2g_HS2_B, GAAGGTTACACAGAACCAGA: 19 extra cells
KlannHS2g_HS2_B, GAAGGTTACACAGAACCAGA: 19 extra cells
KlannHS2g_HS2_B, GGACTATGGGAGGTCACTAA: 28 extra cells
KlannHS2g_HS2_B, GGACTATGGGAGGTCACTAA: 28 extra cells
Klannchr1_HS3, CCATGAGTAGAGGGCAGACA: 26 extra cells
Klannchr1_HS3, CCATGAGTAGAGGGCAGACA: 25 extra cells
Klannchr1_HS4, TACTAGGCTGACTCACTCCA: 64 extra cells
Klannchr1_HS4, TACTAGGCTGACTCACTCCA: 63 extra cells
Number of cells
0
1
5
10
0
2
4
6
8
10
gRNA UMI count (that guide)

Unassigning fishash+'s extra calls: for each guide, the (guide, cell) entries fishash+ assigns and fishash does not are set to unassigned if their UMI count is below a threshold, so those cells become controls for that guide group. Unassigning all of them gives exactly fishash's assignment, since in this situation fishash's assignments are perfectly nested in fishash+'s. "Before" is fishash+ as is, so this shows the change in fishash+'s p-values when those "questionable" calls are removed.

This could suggest that removing the guide-cell pairs with UMI counts of 2 and 3 helps, but there is no evidence that removing just the UMI counts of 2 helps. Very little evidence either way in such a small sample.

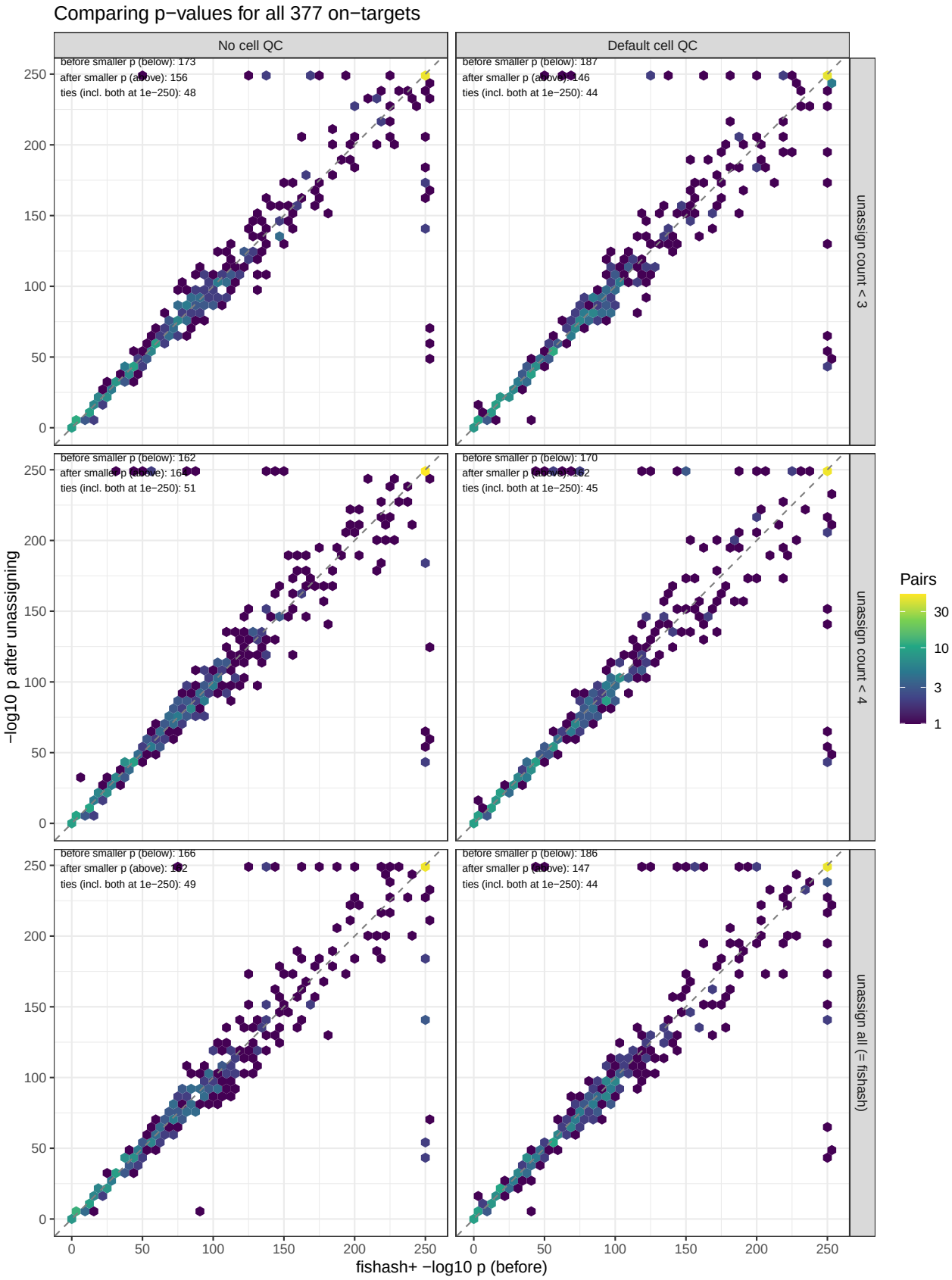

No cell QC
Default cell QC
unassign count < 3
unassign count < 4
unassign all (= fishash)
highlighted pairs:
before smaller p (below): 8
after smaller p (above): 9
ties (incl. both at 1e−250): 1
highlighted pairs:
before smaller p (below): 12
after smaller p (above): 6
ties (incl. both at 1e−250): 0
highlighted pairs:
before smaller p (below): 11
after smaller p (above): 7
ties (incl. both at 1e−250): 0
highlighted pairs:
before smaller p (below): 9
after smaller p (above): 9
ties (incl. both at 1e−250): 0
highlighted pairs:
before smaller p (below): 12
after smaller p (above): 6
ties (incl. both at 1e−250): 0
highlighted pairs:
before smaller p (below): 12
after smaller p (above): 6
ties (incl. both at 1e−250): 0
−log10 p after unassigning
fishash+ −log10 p (before)
0
50
100
150
200
250
Enhancer x HBE1/HBG1/HBG2
Other

Checking the z scores: again, In this case there is no evidence of any change.

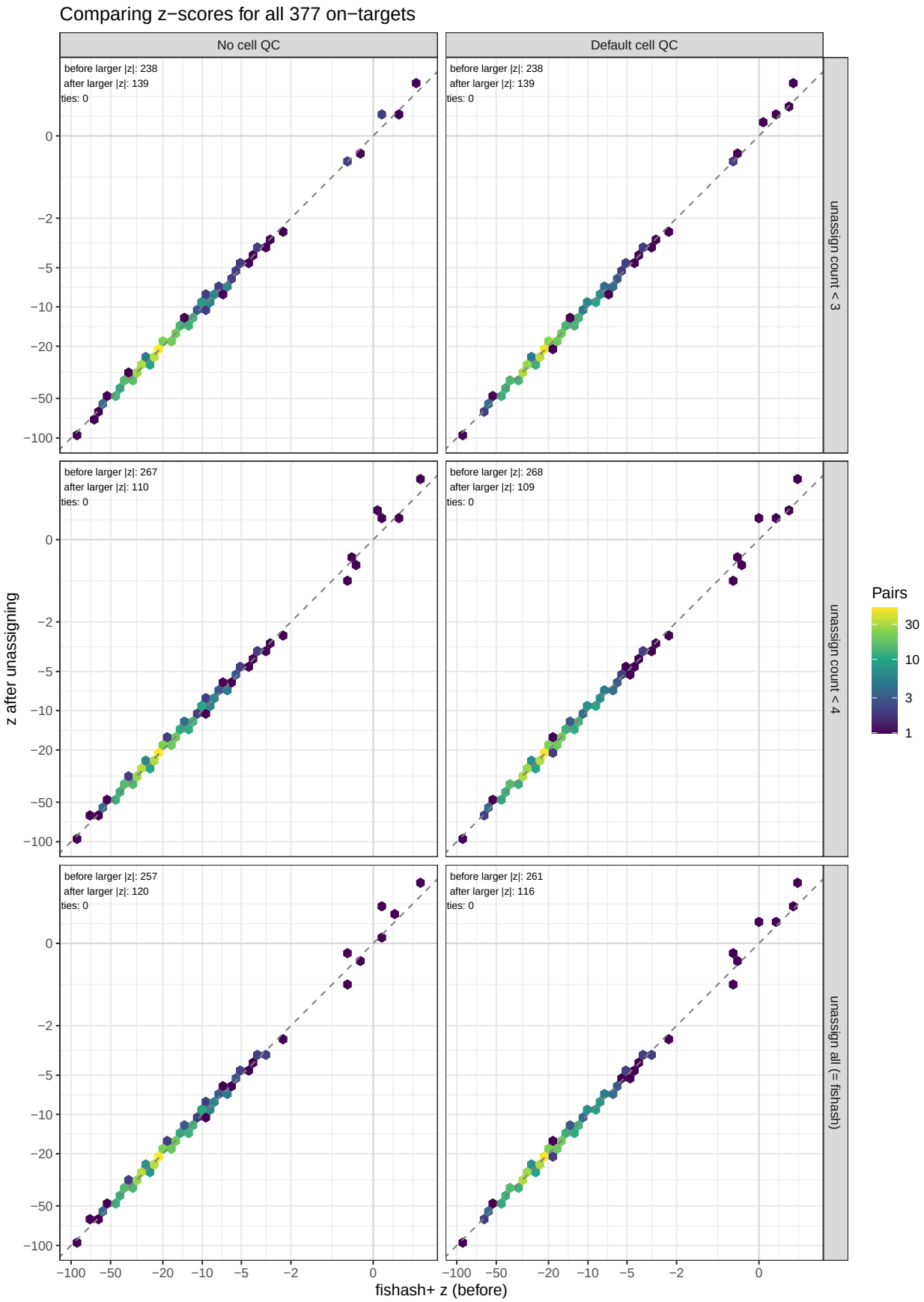

No cell QC
Default cell QC
highlighted pairs:
before larger |z|: 14
after larger |z|: 4
ties: 0
highlighted pairs:
before larger |z|: 13
after larger |z|: 5
ties: 0
unassign count < 3
unassign count < 4
unassign all (= fishash)
z after unassigning
fishash+ z (before)
10
0
−10
−20
Enhancer x HBE1/HBG1/HBG2
Other

The same unassignment experiment on the 377 on-target positive controls. Here too fishash's assignments are nested in fishash+'s for all 754 guides, so unassigning all of fishash+'s extra calls gives exactly fishash's assignment.

Points below the diagonal: fishash+ before unassigning has the smaller p.



On-target z-score analysis.


Comparing z−scores for all 377 on−targets
No cell QC
Default cell QC
unassign count < 3
unassign count < 4
unassign all (= fishash)
before larger |z|: 238
after larger |z|: 139
ties: 0
before larger |z|: 267
after larger |z|: 110
before larger |z|: 268
after larger |z|: 109
before larger |z|: 257
after larger |z|: 120
before larger |z|: 261
after larger |z|: 116
z after unassigning
fishash+ z (before)
0
−2
−5
−10
−20
−50
−100
Pairs
30
10
3
1